Geometric Proofs of Newton's Inverse-Square Law and Kepler's Conic Section Orbit

Abstract

Kepler's laws state that a planet moves on an ellipse with the Sun at a focus (the force center) and sweeps out equal areas in equal times (constant areal speed). We present a fully geometric proof—built from explicit Euclidean straightedge-and-compass constructions—that the acceleration is central and proportional to $1/r^2$ (Newton's universal gravitational law). Our emphasis is on an elementary *Principia*-style argument that avoids differential equations. The proof should be accessible to high-school readers.

1 Introduction

Newton's original geometric arguments connecting Keplerian motion to an inverse-square central force appear in the first edition (1687) of the *Principia* [New87]; see also the 1846 Motte translation [New46]. In a nutshell: from Kepler's laws (first and second laws, both 1609)—knowing in particular that the orbit is an ellipse and that equal areas are swept in equal times—Newton inferred that the acceleration is central and proportional to $1/r^2$.

Modern reconstructions and textbook-style accounts of Newton's ellipse $\Rightarrow 1/r^2$ direction can be found in, e.g., [Cha95, OW12, Mar11]. The reverse direction (inverse-square $\Rightarrow$ conic orbits) is famously associated with Feynman's 1964 "lost lecture" [GG96], with earlier related ideas already appearing in Maxwell's 1877 treatment and related 19th-century work.

The goal of the present paper is not to contribute to the reverse-direction derivation, but to give an alternative, fully geometric route to the inverse-square law from Keplerian (in particular, elliptic) geometry.

Two aspects of the argument:

(i) From Kepler's area law: central direction. From Kepler's second law (1609; equal areas in equal times about the Sun/focus), the acceleration is directed toward the focus (Book I, Proposition 1, 1687).

(ii) From the ellipse shape: inverse-square magnitude. With Kepler's first law (1609; elliptic orbit with the Sun at a focus), the magnitude of the centripetal acceleration scales as $1/r^2$ (Book I, Propositions 11–13, 1687, in modern numbering/conventions).

We briefly summarize what the main cited references emphasize, to clarify how the present proof fits among them (we refer to the bibliography for full details). For historical discussion of how Book I evolves across different versions/editions, see also [Nov].

- Newton (1687) [New87]. Newton's masterpiece uses synthetic (polygonal) arguments together with an early notion of limits (ultimate/last ratios): equal areas in equal times imply a central impulse toward the focus, and the inverse-square dependence is extracted via geometric properties of the ellipse.

- Chandrasekhar (1995) [Cha95]. A detailed modern reconstruction of Newton's arguments, combining synthetic geometry with contemporary analytic language.
- Goodstein & Goodstein (1996) [GG96]. A geometric narrative inspired by Feynman's 1964 "lost lecture," emphasizing a nearly calculus-free route for the reverse direction (inverse-square $\Rightarrow$ conic orbits) using hodograph synthesis.
- Derbes (2001) [Der01]. A clear pedagogical account of hodographic (velocity-space) methods for the Kepler problem, including the parabolic case.
- Stavek (2019) [Sta19]. A geometric exploration centered on Newton's parabola and related classical constructions (directrix, pedal curve, evolute, subtangent/subnormal, and a Ptolemy-circle/hodograph viewpoint).
- Markowsky (2011) [Mar11]. A careful retelling and streamlining of Newton's arguments, with an emphasis on geometric structure and pedagogy. Markowsky also emphasizes that whether Newton regarded "having a solution" as sufficient for the reverse direction (inverse-square $\Rightarrow$ conics) is debatable: one can justify the step via existence/uniqueness results for the associated ordinary differential equations. It also seeks a *Principia*-style proof of energy conservation, from which one can derive that the trajectory is a conic.
- Provost & Bracco (2008) [PB08]. A hybrid yet insightful and concise derivation highlighting conserved quantities—essentially the Laplace–Runge–Lenz vector—that analytical approaches often use to show that the path is a conic. This paper offers an elementary view based on areal speed and related geometric relations, and also summarizes Jakob Hermann's early-18th-century proof (c. 1710s) and the aforementioned proof of Hamilton (1847).
- Woan [Woa06]. A standard mechanics approach in polar coordinates, deriving the conic form by solving for $r(\theta)$ using angular momentum and energy, in the tradition of analytic approaches developed by mathematicians such as Bernoulli (late 17th to early 18th century).
- Simha (2021) [Sim21]. A calculus-light, trigonometric presentation of Kepler's first law; it substantially overlaps with the Markowsky-style approach, possibly without awareness of that earlier source. It is mentioned here because it uses the law of cosines effectively.
- van Haandel & Heckman (2009) [vHH09]. A freshman-oriented teaching note emphasizing geometric intuition. Besides giving a quick account of Feynman and Newton's methods, it also provides a geometric construction of the LRL vector that offers further insight into the reverse problem.

In Sections 2–3 we briefly repeat Newton's argument (1687, with his figure) showing why the area theorem implies a centripetal (central) force. In Sections 4–8 we present a geometric method, based on the auxiliary circle and an affine map, that yields the inverse-square dependence. The proof here is closer in spirit to Newton's synthetic geometry, but it is organized around a particular auxiliary construction: we compute the short-time "departure from the tangent" on the ellipse via an explicit affine map to the auxiliary circle, show that the corresponding quantities match, and then the rest follows.

2 Setup and assumptions

Let an ellipse have semi-major axis a, semi-minor axis b, and foci F and F'. A point mass m moves along the ellipse in a plane. Let $r = |FA|$ be the instantaneous distance to the focus F. See Figure 3 for the auxiliary-circle/ellipse construction used later.

Hypothesis H1 (ellipse): the orbit is an ellipse with focus at F.

Hypothesis H2 (area law): *equal areas are swept in equal times* about F (Kepler's second law, 1609). Equivalently, there is a constant k such that for any time interval of length t, the swept area satisfies

$$S = k\,t. \tag{2.1}$$

It is convenient to write $L := 2k$ (twice the constant area rate), so that the swept area in time t is $S = (L/2)t$.

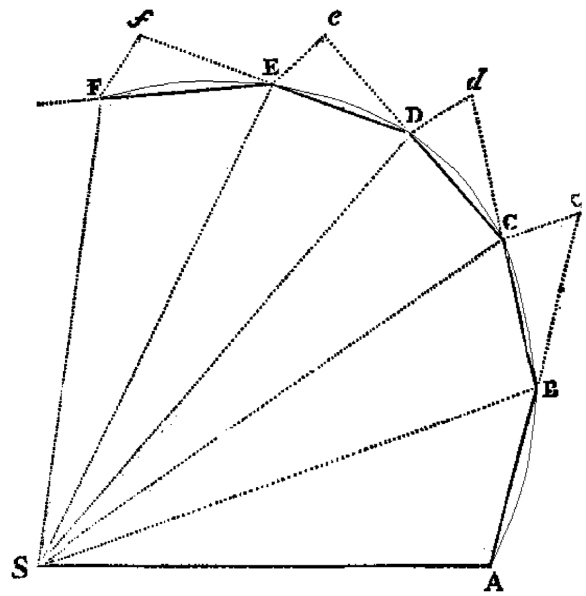

Figure 1: Newton's area theorem construction, reproduced from [New87]. Because it is an original 1687 diagram, it naturally looks coarse/ancient, but the geometric idea is clear.

In modern mechanics, this constant is the *specific angular momentum* (angular momentum per unit mass); many texts denote it by h (see, e.g., [Cha95, GG96]). The ordinary angular momentum is then $\mathcal{L} = mL$.

We assume the motion is smooth enough that "short time step" expansions are valid in the usual limiting sense (errors are higher order in the small time step).

3 Central direction from constant areal speed

3.1 Newton's area-law centripetal direction

We briefly recall Newton's original geometric implication of Kepler's area law (1609) in the *Principia* (Book I, Proposition 1, 1687) [New87]. We do not repeat Newton's elementary geometric proof here; an accessible modern retelling is in Section 22 ("The Area Theorem") of Chapter 5 of [Cha95]. A short online exposition in the same geometric spirit is also available in [Non], and a detailed fill-in of some steps in Newton's Proposition 1 is given in [Nau03].

Approximate the orbit by a broken line $ABCDEF\cdots$ traversed in equal time steps, with instantaneous "impulses" at the vertices. Fix the point S about which areas are swept. If there is no impulse at B, then after AB the motion would continue straight to a point c on the extension of AB with $Bc = AB$. Newton's construction draws through c a line parallel to SB and places C on this line so that $C \in SC$ (see Figure 1); this makes the successive swept triangles $\triangle SAB$ and $\triangle SBC$ have equal areas. Conversely, if equal areas are swept in equal times, then at each vertex the "jump" from the straight continuation (e.g., from c back to C) must lie along BS (and similarly at $C, D, \ldots$). In the smooth limit, the acceleration is therefore always directed along AS, i.e., the force is *centripetal* with center S.

Thus it remains to determine the magnitude of the centripetal acceleration $a(r)$.

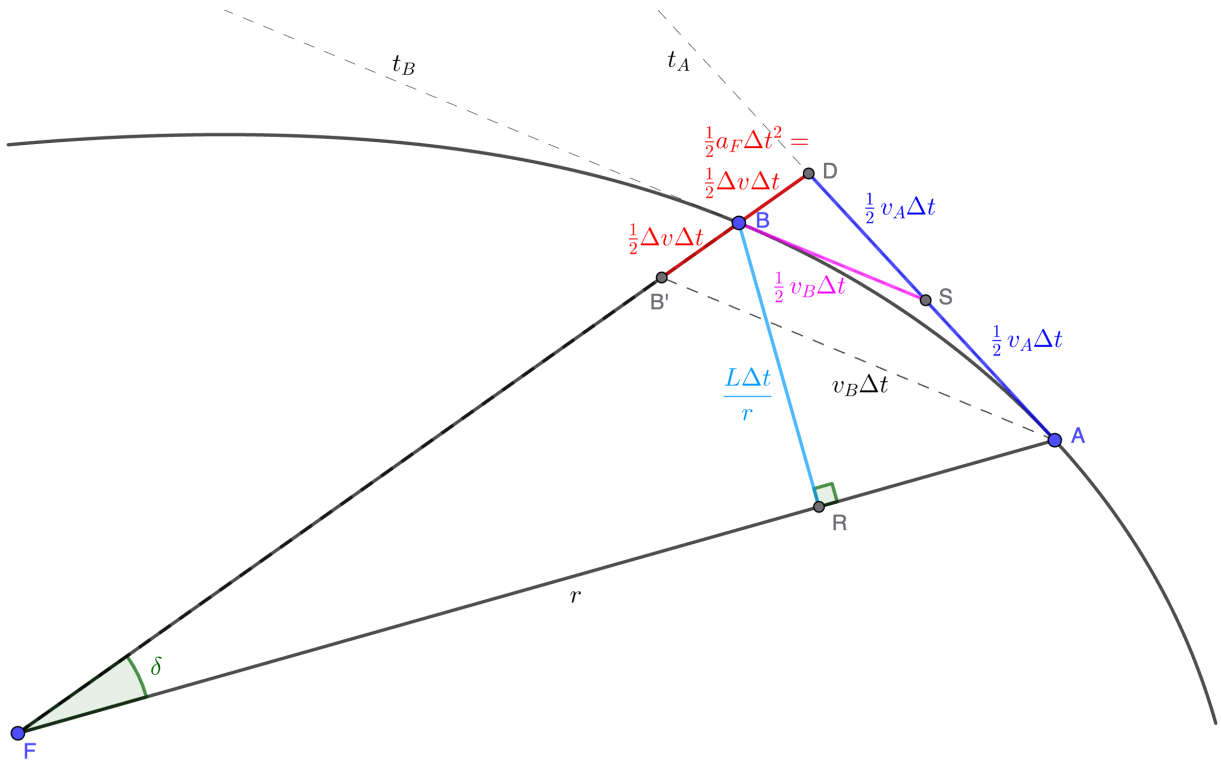

Figure 2: Local construction used in Theorems 3.1 and 3.2.

3.2 Local distance relationships for BD and BR^2

Lemma 3.1 (Deflection identity)**.** *Over an evanescent time step Δt, the deflection produced by the mean acceleration a_F satisfies*

$$BD = \frac{1}{2}a_F(\Delta t)^2 = \frac{1}{2}\Delta v\,\Delta t, \qquad (\Delta v = a_F\Delta t). \tag{3.1}$$

With the tangent construction, S bisects AD; and in the limit $B \to A$,

$$AS = SD = SB = \frac{1}{2}\,\widehat{AB}. \tag{3.2}$$

Proof. From $\Delta v = a_F\Delta t$,

$$\frac{1}{2}a_F(\Delta t)^2 = \frac{1}{2}(a_F\Delta t)\Delta t = \frac{1}{2}\Delta v\,\Delta t.$$

Thus the deflection in time Δt equals the distance generated by the mean velocity increment $\frac{1}{2}\Delta v$ during Δt, namely the constructed segment BD. By construction, S is the midpoint of AD, so S bisects AD. As $B \to A$, the chordal midpoints coalesce in the ultimate ratio, giving

$$AS = SD = SB = \frac{1}{2}\,\widehat{AB}.$$

□

Lemma 3.2 (Ratio BD/BR^2 and inverse-square form)**.** *Let F be the center of force, $r = FB$, and $BR \perp FB$. Let L be twice the areal speed, so*

$$\Delta A = \frac{L}{2}\Delta t.$$

Then, as $\Delta t \to 0$,

$$BR \sim \frac{L\Delta t}{r}, \qquad \frac{BD}{BR^2} \sim \frac{1}{2}\, a_F\, \frac{r^2}{L^2}. \tag{3.3}$$

Hence if $\dfrac{BD}{BR^2} \to c$, *then*

$$a_F = \frac{2cL^2}{r^2}, \qquad \vec{a}_F = -\frac{2cL^2}{r^2}\,\hat{r}. \tag{3.4}$$

Proof. From Prop. I (1846), swept areas are proportional to times; for an evanescent step,

$$\Delta A \sim \frac{1}{2} r\, BR.$$

Using $\Delta A = \frac{L}{2}\Delta t$ gives

$$\frac{L}{2}\Delta t \sim \frac{1}{2} r\, BR \quad \Longrightarrow \quad BR \sim \frac{L\Delta t}{r}.$$

By Theorem 3.1,

$$BD = \frac{1}{2} a_F (\Delta t)^2.$$

Substitute $BR \sim L\Delta t/r$ to obtain

$$\frac{BD}{BR^2} \sim \frac{\frac{1}{2} a_F (\Delta t)^2}{(L\Delta t/r)^2} = \frac{1}{2}\, a_F\, \frac{r^2}{L^2}.$$

So if the ratio tends to c, then $a_F = 2cL^2/r^2$, directed toward F, i.e.

$$\vec{a}_F = -\frac{2cL^2}{r^2}\,\hat{r}.$$

□

4 Auxiliary circle and the affine map

Let ℓ be the major axis and let O be the center of the ellipse. Recall that the ellipse has semi-major axis a and semi-minor axis b. Consider the *major-axis circle* $\mathcal{C}$: the circle centered at O with radius a (so its diameter equals the major axis length $2a$). This is also the standard *auxiliary circle* associated to the ellipse.

Define the affine map Φ geometrically as follows. For any point P in the plane, let P_T be the foot of the perpendicular from P to ℓ. On the same perpendicular line $P_T P$ choose a point P_0 on the same side of ℓ as P such that

$$P_T P_0 = \frac{b}{a}\, P_T P. \tag{4.1}$$

Then set $\Phi(P) := P_0$.

Equivalently, Φ fixes ℓ pointwise and scales all lengths perpendicular to ℓ by the constant factor b/a. This map sends lines to lines, preserves parallelism, and scales all areas by b/a. In the standard auxiliary circle construction, Φ maps $\mathcal{C}$ to the ellipse.

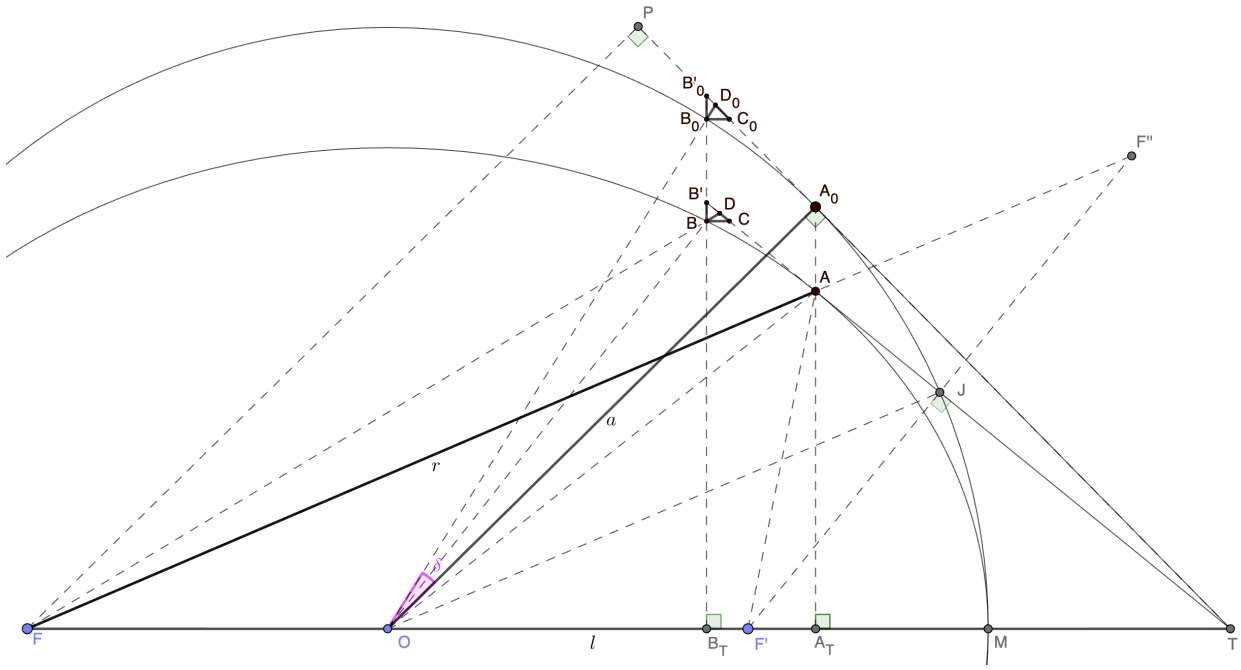

Figure 3: Affine relation of motion between the auxiliary circle and the ellipse.

Tangents via secants (Newton's 1687 viewpoint). Newton often treats a tangent as the *ultimate position* of a secant: if A is a point on the curve and B is a nearby point, then the chord line AB approaches the tangent line at A as $B \to A$ (in Newton's 1687 language, as one takes the "ultimate ratio"). In our setting this provides an alternative justification for the fact that Φ carries circle tangents to ellipse tangents.

Indeed, let $A_0 \in \mathcal{C}$ correspond to $A = \Phi(A_0)$ on the ellipse, and let $B_0 \in \mathcal{C}$ correspond to $B = \Phi(B_0)$. Since Φ is affine, it maps the secant (chord) line A_0B_0 to the secant line AB, and it preserves incidences and parallelism. As $B_0 \to A_0$, the secant line A_0B_0 approaches the tangent to $\mathcal{C}$ at A_0, while AB approaches the tangent to the ellipse at A. Therefore, in the same limiting (secant-to-tangent) sense, Φ maps the circle tangent at A_0 to the ellipse tangent at A.

A few simple geometric consequences we use repeatedly:

- Straight lines map to straight lines under Φ.
- Horizontal lengths (parallel to ℓ) are unchanged by Φ.
- Vertical lengths (perpendicular to ℓ) scale by b/a.
- Parallel lines stay parallel under Φ, and ratios of lengths along such lines are unchanged by Φ.
- Areas scale by b/a.
- The inverse map Φ^{-1} sends $\Phi(A)$ back to A and scales vertical lengths by a/b.

Even though Newton (1687) did not describe affine transformations explicitly, he was well aware of these properties and applied them throughout his books. He also used clever polygonal approximations to treat areas under curves (a method far ahead of its time).

5 Tangent transfer and matching normal drops

We record a concrete straightedge-and-compass configuration that makes the affine "transport" of the short-time normal departure from the tangent completely explicit. The purpose of this section is to obtain the geometric local-drop estimate; the kinematic conversion to acceleration then uses Theorems 3.1 and 3.2 from Section 3.

5.1 Circle companions and tangent transfer

Let ℓ be the major axis, and let Φ map the auxiliary circle $\mathcal{C}$ to the ellipse. From A and B drop perpendiculars to ℓ and extend these same vertical lines to meet the auxiliary circle $\mathcal{C}$ at A_0 and B_0 (choose the intersections on the same side of ℓ as A and B). By construction, $\Phi(A_0) = A$ and $\Phi(B_0) = B$. Draw the tangent to $\mathcal{C}$ at A_0 and let it meet ℓ at T.

Lemma 5.1 (Tangent transfer with fixed T)**.** *Under Φ, the circle tangent at A_0 maps to the ellipse tangent at A. Moreover, since Φ fixes ℓ pointwise, the intersection point $T \in \ell$ is the same for both tangents.*

Proof. TA_0 is the tangent to the circle at A_0. For points B_0 on the circle taken sufficiently near A_0, all such B_0 lie on the same side of the line TA_0.

The affine bijection Φ sends lines to lines and half-planes to half-planes, so it preserves the relation "lying on the same side of a line." Hence the line $\Phi(T)\Phi(A_0)$ meets the ellipse at $A = \Phi(A_0)$. Since $\Phi(T) = T$ and all nearby image points $B = \Phi(B_0)$ lie on one side of TA, the line TA is tangent to the ellipse at A.

Here we implicitly use the fact that a conic is smooth and therefore has a unique and well-defined tangent line at each point. □

5.2 The B, C, D construction (ellipse and circle)

Define C as the intersection of the ellipse tangent TA with the line through B parallel to ℓ, and define $D := TA \cap FB$. Likewise, on the circle tangent TA_0 define C_0 as the intersection with the line through B_0 parallel to ℓ, and define $D_0 := TA_0 \cap OB_0$. By construction, $BC \parallel \ell$ and $B_0C_0 \parallel \ell$.

5.3 The point J is on $\mathcal{C}$

Extend the ray AF beyond A and mark a point F'' on this ray such that $AF'' = AF'$ (so $\triangle AF'F''$ is isosceles). Let $J := AT \cap F'F''$.

Lemma 5.2 (Ellipse tangent bisects the focal angle)**.** *The tangent line AT bisects $\angle F''AF'$.*

In particular (by the isosceles geometry), $AT \perp F'F''$ and J is the midpoint of $F'F''$. But we also know O bisects FF', so $OJ \parallel FA$ and $OJ = FF''/2 = a$, which proves that J lies on the circle $\mathcal{C}$.

5.4 Matching the normal drops

The triangle similarity relations in the tangent line through T give

$$B_0D_0 = B_0C_0\,\frac{OD_0}{OT}, \qquad BD = BC\,\frac{FD}{FT}. \tag{5.1}$$

Dividing and using that $\Phi(B_0) = B$ and Φ maps the line through B_0 parallel to ℓ to the line through B parallel to ℓ (since Φ fixes ℓ and preserves parallelism), while also mapping the tangent line TA_0 to TA, we have $\Phi(C_0) = C$. Hence the horizontal segment B_0C_0 is carried to BC, and since Φ preserves horizontal lengths, $B_0C_0 = BC$.

$$\frac{B_0D_0}{BD} = \frac{OD_0}{OT} \cdot \frac{FT}{FD}. \tag{5.2}$$

Letting $B \to A$ (so $D_0 \to A_0$ and $D \to A$), we obtain

$$\lim_{B \to A} \frac{B_0D_0}{BD} = \frac{OA_0}{OT} \cdot \frac{FT}{FA}. \tag{5.3}$$

Lemma 5.3 (Matching drops)**.** *In the infinitesimal-step limit, the circle and ellipse normal drops from the tangent agree:*

$$\boxed{\lim_{B \to A} \frac{B_0D_0}{BD} = 1.} \tag{5.4}$$

Proof. Now $F, O, T \in \ell$, and by construction $J \in AT$ and $OJ \parallel FA$. Hence triangles $\triangle TFA$ and $\triangle TOJ$ are similar. Therefore

$$\frac{FT}{FA} = \frac{OT}{OJ}.$$

Since $A_0, J \in \mathcal{C}$ we have $OA_0 = OJ = a$, and consequently the right side of (5)

$$\frac{OA_0}{OT} \cdot \frac{FT}{FA} = \frac{a}{OT} \cdot \frac{OT}{a} = 1.$$

which proves (6). □

5.5 Transporting swept area from the ellipse to the circle

Let A be the current point on the ellipse and B the position after a small time increment t. Let the swept area about the focus be

$$S = \text{area}(\triangle FAB) = \frac{L}{2}\, t. \tag{5.5}$$

We compare this to a corresponding small triangle on the circle. Let

$$S_0 := \text{area}(\triangle OA_0B_0), \tag{5.6}$$

Two geometric scalings relate S and S_0:

1. Height scaling from O to F. In the construction used in our discussion, the line through F parallel to a fixed circle radius implies that the perpendicular height from F to the chord AB is (r/a) times the perpendicular height from O to the same chord direction. Hence

$$\text{area}(\triangle FAB) = \frac{r}{a}\, \text{area}(\triangle OAB). \tag{5.7}$$

2. Affine area scaling. Since Φ scales area by b/a,

$$\text{area}(\triangle OAB) = \frac{b}{a}\, \text{area}(\triangle OA_0B_0) = \frac{b}{a}\, S_0. \tag{5.8}$$

Combining,

$$S = \text{area}(\triangle FAB) = \frac{r}{a} \cdot \frac{b}{a}\, S_0 = \frac{rb}{a^2}\, S_0, \tag{5.9}$$

so

$$S_0 = \frac{a^2}{rb}\, S = \frac{a^2}{rb} \cdot \frac{L}{2}\, t = \frac{La^2}{2br}\, t. \tag{5.10}$$

5.6 Circle tangent and sagitta

As $B_0 \to A_0$, (5.10) gives

$$A_0D_0 = \frac{2S_0}{a} = \frac{2}{a} \cdot \frac{La^2}{2br}\, t = \frac{La}{br}\, t. \tag{5.11}$$

Now use elementary circle geometry relating the *normal drop* from the circle to its tangent (the sagitta). For a circle of radius a, the perpendicular distance from the circle point reached to the tangent line satisfies (to leading order)

$$B_0D_0 \approx \frac{(A_0D_0)^2}{2a}. \tag{5.12}$$

Substituting (5.11) gives

$$B_0D_0 \approx \frac{1}{2a}\left(\frac{La}{br}t\right)^2 = \frac{L^2a}{2b^2r^2}\, t^2. \tag{5.13}$$

5.7 Proof 1 (ellipse via affine transported drops)

Proposition 5.4 (Proof 1: direct problem for the ellipse)**.** *Under Hypotheses H1–H2, the acceleration is focus-directed and has inverse-square magnitude:*

$$\mathbf{a}_F(r) = -\mu\, \frac{\mathbf{r}}{r^3}, \qquad \mu = \frac{L^2a}{b^2} = \frac{L^2}{p}, \tag{5.14}$$

where $p = b^2/a$ is the semi-latus rectum and $\mathbf{r}$ is the position vector from F.

Proof. From Theorem 5.3 and the sagitta estimate above,

$$BD \sim B_0D_0 \sim \frac{L^2a}{2b^2r^2}\, t^2.$$

Also, by the area-law relation (equivalently, the first asymptotic in Theorem 3.2),

$$BR \sim \frac{L}{r}\, t.$$

Therefore

$$\frac{BD}{BR^2} \to \frac{a}{2b^2} = \frac{1}{2p}, \qquad \left(p = \frac{b^2}{a}\right).$$

Applying Theorem 3.2 with $c = \frac{1}{2p}$ gives

$$a_F = \frac{2cL^2}{r^2} = \frac{L^2}{p} \cdot \frac{1}{r^2} = \frac{L^2a}{b^2} \cdot \frac{1}{r^2}.$$

Combining this magnitude with Section 3's central-direction conclusion yields

$$\mathbf{a}_F(r) = -\mu\, \frac{\mathbf{r}}{r^3}, \qquad \mu = \frac{L^2a}{b^2} = \frac{L^2}{p}.$$

For the areal constant k with $L = 2k$, this is equivalently $\mu = 4k^2/p$. □

6 Alternative auxiliary-circle proof (same inverse-square law)

In this section we give a second derivation of the same $1/r^2$ law, still using the auxiliary circle C (center O, radius a), but emphasizing a different set of local ellipse facts so that the argument has as few moving parts as possible.

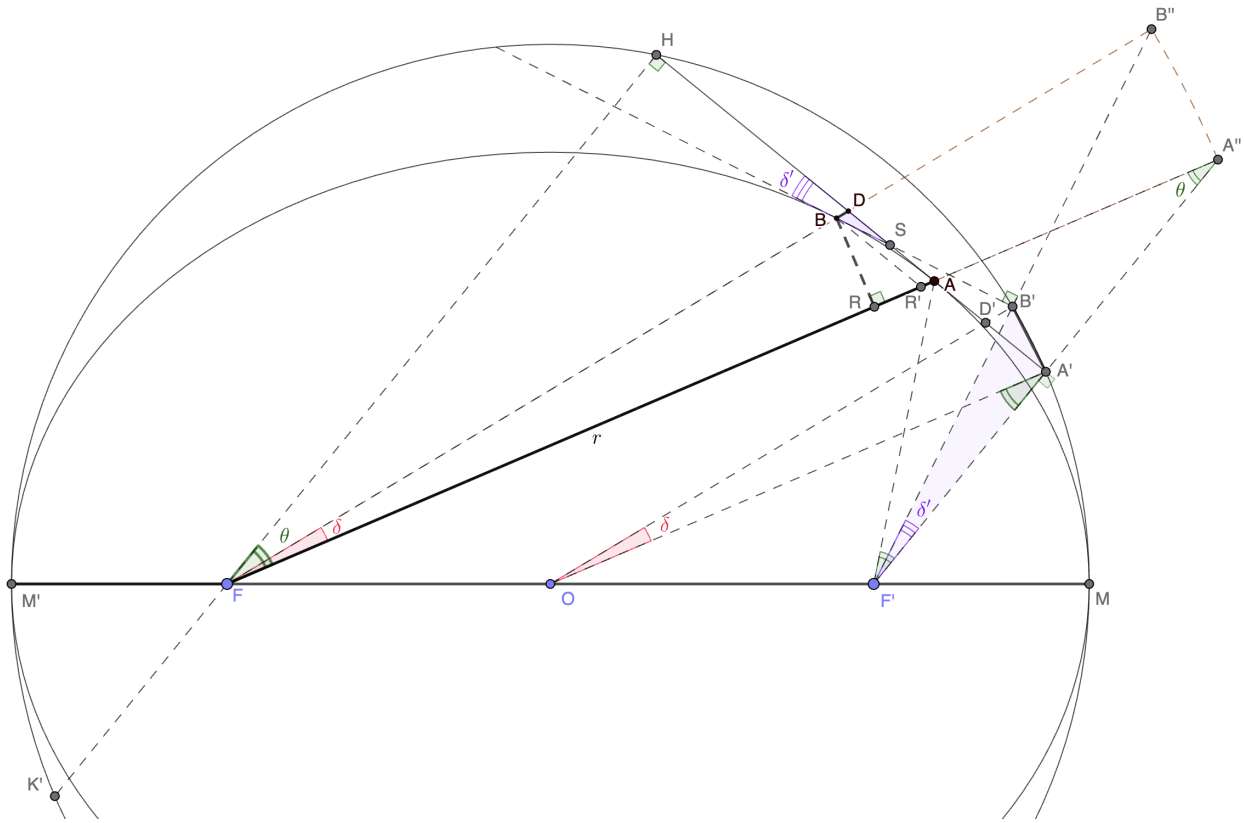

Figure 4: Alternative auxiliary-circle construction used in Section 6.

6.1 Minimal setup

Let the ellipse have foci F, F'. Fix a point A on the ellipse and write $r = FA$. Let the tangent at A meet the auxiliary circle again at A'. For a nearby point B on the ellipse, let its tangent meet the same auxiliary circle at B'. We let $B \to A$. Denote these tangent lines by t_A and t_B, respectively; the instantaneous velocity vectors $\mathbf{v}_A$ and $\mathbf{v}_B$ are directed along t_A and t_B.

Let D be the intersection point on the tangent construction (as in Figure 4), and let R be the foot used to form the small segment BR (the small transverse piece in the right triangle at R).

6.2 Two standalone lemmas

Lemma 6.1 (Product identity)**.** *With the notation of Figure 4,*

$$FH \cdot F'A' = (a - c)(a + c) = b^2, \tag{6.1}$$

where $c = OF$ is the focal distance (so $a^2 = b^2 + c^2$).

Proof. Since $FK' \parallel F'A'$ in the construction, triangles with corresponding sides along these parallels are similar, and the segment on the parallel through F has the same length as the corresponding segment on the tangent through F'. In particular,

$$F'A' = FK'. \tag{6.2}$$

Therefore

$$FH \cdot F'A' = FH \cdot FK' = (a-c)(a+c) = a^2 - c^2 = b^2, \tag{6.3}$$

which is the standard ellipse identity. □

Lemma 6.2 (Newton small-distance lemma)**.** *As $B \to A$, the little segments cut off by nearby tangents/secants agree to first order; thus the similar-triangle relations used below are valid in the Newtonian (1687) "ultimate ratio" sense (any error is higher order and vanishes in the limit).*

6.3 Proof 2 (constant $BD/BR^2 \Rightarrow$ inverse-square law)

Step 1: a similarity formula for BD. As $B \to A$, similarity of the indicated triangles in Figure 4 gives

$$\triangle SDB \sim \triangle F'A'B' \qquad \Longrightarrow \qquad BD = \frac{1}{2}\, AD \cdot \frac{A'B'}{A'F'}. \tag{6.4}$$

(The factor 1/2 is the same midpoint/evanescent factor as in the earlier Newton-style tangent-departure setup.)

In particular, this step highlights again the pair of similar triangles $\triangle F'A'B'$ and $\triangle SDB$ that drives the computation.

Step 2: express both AD and $A'B'$ in terms of BR. From the focal-geometry similarity in the figure,

$$AD = \frac{AF}{FH}\, BR = \frac{r}{FH}\, BR. \tag{6.5}$$

From the auxiliary-circle similarity (equivalently, the affine scaling ellipse $\leftrightarrow$ circle),

$$A'B' = \frac{OA'}{F'A}\, BR = \frac{a}{r}\, BR. \tag{6.6}$$

Step 3: multiply and collapse. Substituting the two expressions into the BD formula yields

$$BD = \frac{1}{2}\left(\frac{r}{FH} BR\right)\left(\frac{a}{r} BR\right)\frac{1}{A'F'} \tag{6.7}$$

$$= \frac{1}{2}\, a\, \frac{BR^2}{FH \cdot A'F'}. \tag{6.8}$$

Using Theorem 6.1 (so that $FH \cdot A'F' = b^2$) we obtain

$$BD = \frac{1}{2}\, a\, \frac{BR^2}{b^2}. \tag{6.9}$$

Finally, using the standard ellipse identity $p = b^2/a$ (the semi-latus rectum), one may rewrite the above in the equivalent form

$$BD = \frac{1}{2p}\, BR^2, \tag{6.10}$$

so

$$\frac{BD}{BR^2} \to \frac{1}{2p}.$$

Applying Theorem 3.2 with $c = \frac{1}{2p}$ gives immediately

$$a_F = \frac{2cL^2}{r^2} = \frac{L^2}{p}\frac{1}{r^2}.$$

□

6.4 Proof 3 (hodograph-style computation)

From Theorem 6.1,

$$F'A' = \frac{b^2}{FH}. \tag{6.11}$$

Let $v = |\mathbf{v}|$ be the speed at A. Since the velocity direction is tangent to the orbit and FH is perpendicular to that tangent (as in the construction), the areal-rate constant gives

$$FH \cdot v = L. \tag{6.12}$$

Combining,

$$F'A' = \frac{b^2}{FH} = \frac{v\, b^2}{L}. \tag{6.13}$$

For two nearby points A, B we therefore have, in the Newtonian small-step sense,

$$A'B' = \Delta(F'A') = \frac{b^2}{L}\,\Delta v = \frac{b^2}{L}\,a_F\,\Delta t, \tag{6.14}$$

where a_F is the magnitude of the (focus-directed) acceleration and we used Theorem 3.1 ($\Delta v = a_F \Delta t$). On the other hand, from the auxiliary-circle similarity used above,

$$A'B' = \frac{OA'}{F'A}\,BR = \frac{a}{r}\,BR. \tag{6.15}$$

Eliminating $A'B'$ gives

$$a_F = \frac{L}{b^2}\,\frac{A'B'}{\Delta t}. \tag{6.16}$$

Next, using again the auxiliary-circle similarity $A'B' = \frac{a}{r}BR$ and Theorem 3.2 ($BR \sim L\Delta t/r$), we have

$$A'B' = \frac{a}{r}\,\frac{L\,\Delta t}{r} = \frac{a}{r^2}\,L\,\Delta t. \tag{6.17}$$

Combining equations (6.18)–(6.19) yields

$$a_F = \frac{L^2}{p}\,\frac{1}{r^2}. \tag{6.18}$$

exactly the same inverse-square formula as in Theorem 5.4. □

6.5 Discussion: directrix circle and a hodograph viewpoint

Directrix circle variant. One can re-run essentially the same similarity argument by replacing the auxiliary circle $\mathcal{C}$ (center O, radius a) with the *directrix circle* centered at F of radius $2a$ (cf. the construction used in Section 5). Extending FA to meet this directrix circle at A'', one has $A''A'F'$ colinear; similarly, extending FB meets the directrix circle at B'' with $B''B'F'$ colinear. Moreover, A' and B' are midpoints of the segments $A''F'$ and $B''F'$, respectively, so

$$A''B'' = 2\,A'B'. \tag{6.19}$$

Consequently,

$$\triangle A'B'F' \sim \triangle A''B''F', \tag{6.20}$$

and the same chain of similar-triangle identities yields the same ultimate-ratio estimate for the normal drop BD, hence the same inverse-square dependence.

Connection to Maxwell's hodograph. The product identity in Theorem 6.1 (already implicit in classical ellipse geometry) also appears in Maxwell's 1877 discussion of the *hodograph* of planetary motion: for an inverse-square central force the velocity vector $\mathbf{v}(t)$ traces a circle in velocity space (Hamilton's 1847 hodograph), and Maxwell notes that this hodograph circle is similar to the directrix circle, with its "speed origin" at F', but rotated by 90° counterclockwise [Max77].

From this point of view, our construction may be read as relating the geometric drop BD directly to a rotated hodograph circle C. This suggests a third, very short proof that reproduces the same acceleration formula.

7 Other conic sections

Although we have focused on the elliptic (bound) case, the same inverse-square central-force setting also admits other conic-section trajectories. Besides ellipses, the remaining possibilities are the parabola (the threshold case) and the hyperbola (the unbound case), together with a degenerate limiting case discussed next.

7.1 Rectilinear motion as a degenerate conic limit

This extreme limit corresponds to rectilinear (radial) motion, which may be viewed as a degenerate conic with zero areal speed about the force center. In this paper we treat it only as a boundary case and do not analyze it further. Newton does discuss this case in the *Principia*, but it is outside our present scope.

7.2 Hyperbola

For an inverse-square central force, hyperbolic trajectories correspond to unbound motion: the point mass comes in from infinity and escapes back to infinity, with the force center located at a focus of the hyperbola.

See Figure 4 for the auxiliary-circle construction used to carry out the same inverse-square argument in the hyperbola case. In this hyperbola setup, one may again view the auxiliary circle $\mathcal{C}$ as providing a convenient geometric proxy for the (circular) hodograph, with F' taken as the velocity origin. As in the elliptic case, this proxy picture must be rotated by 90° to match the correct direction of the velocity vector in the hodograph construction.

Likewise, the directrix circle ${\mathcal{C}'}_F$ may be regarded as a $2\times$ scaling of the auxiliary circle $\mathcal{C}$ with scaling center F'; consequently it can also be interpreted as a scaled version of the hodograph, again up to the same 90° rotation.

The same key product identity (Theorem 6.1) also holds in this hyperbola setting, though processed differently:

$$F'A' \cdot FH = FH \cdot FK = FM' \cdot FM = (c-a)(c+a) = c^2 - a^2 = b^2, \tag{7.1}$$

using the same tangent/intercept relations (here $c = OF$ and $a^2 = b^2 + c^2$).

Similarly, in the analogue of Proof 2 the computation is again driven by the same pair of similar triangles $\triangle F'A'B'$ and $\triangle SDB$.

There is also a "swapped-focus" situation leading to the same right-branch hyperbola: instead of an attractive (centripetal) force toward F, one may consider a centrifugal force directed away from F' acting on the particle as it moves from A to B. In that case to carry out the proof, the roles of F and F' are interchanged, and the point R is dropped onto the extension of the new line FA. The points H and A' also exchange roles, B' moves to the opposite side of the auxiliary circle $\mathcal{C}$, and D becomes the intersection of the tangent line at A

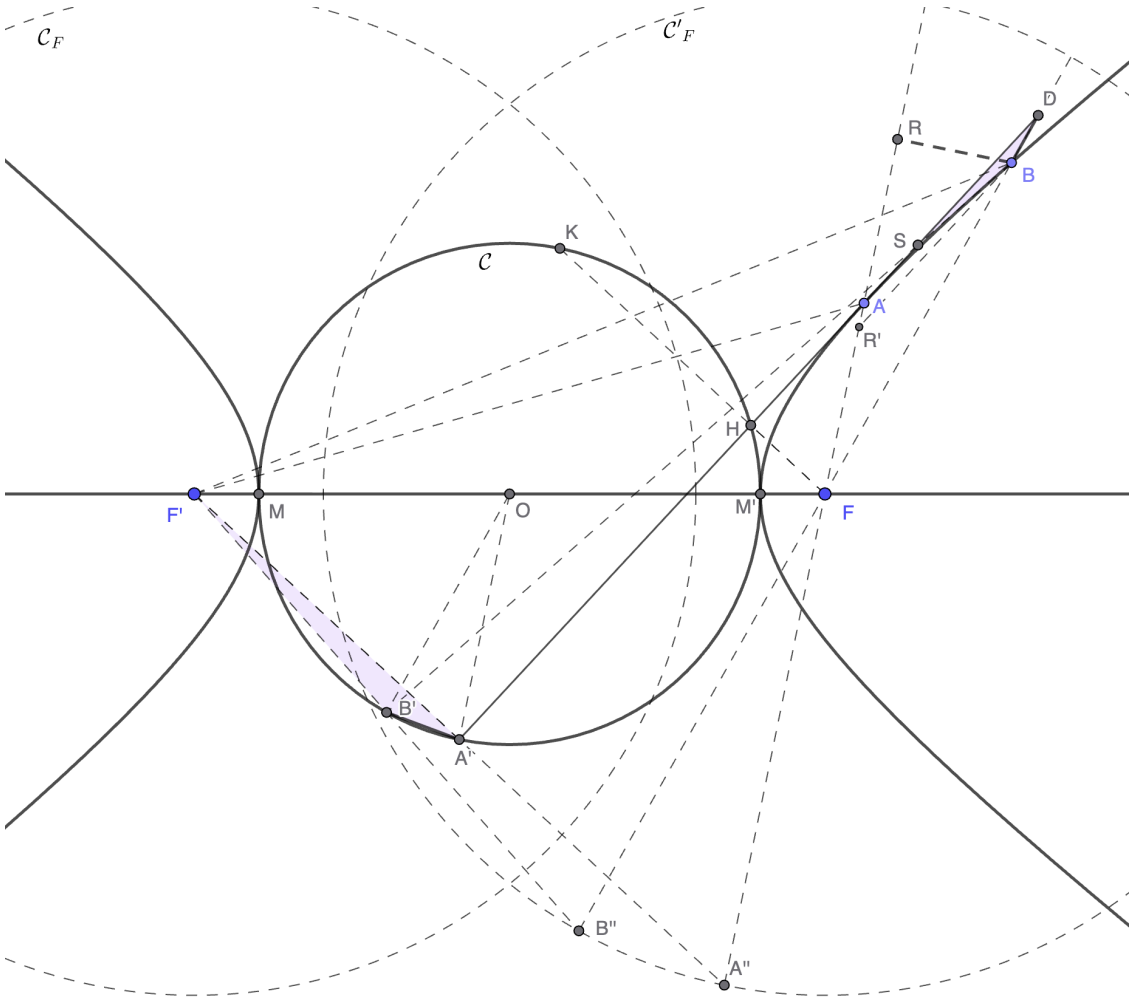

Figure 5: Auxiliary-circle construction for the hyperbola case.

with the new line FB (namely, $F'B$ as in Figure 4). Readers should be able to construct the modified plot accordingly without difficulty, and the proof then follows in exactly the same way.

So once Figure 4 is in place, the remaining steps corresponding to our Proofs 2 and 3 for the hyperbola follow with only minor, mostly notational, changes. In particular, once the hyperbola geometry yields a constant limit for BD/BR^2, Theorem 3.2 gives the inverse-square form directly.

7.3 Parabola

7.3.1 Parabola as a limiting conic and hodograph setup

The parabolic trajectory is the boundary between bound (elliptic) and unbound (hyperbolic) motion. It is also distinguished geometrically as the conic section obtained when the cutting plane is parallel to a generating edge of the cone.

One can also view a parabola as a limiting ellipse in which $a, c \to \infty$ while $(a-c)$ stays finite. Indeed, since $p = b^2/a = (a+c)(a-c)/a$, in this limit one has $p \to 2(a-c)$. It is therefore natural to parametrize the parabola by setting $a - c$ as a single parameter a, so that $p = 2a$. From this perspective, the auxiliary circle

degenerates to the tangent line at the vertex of the parabola, perpendicular to the symmetry axis AF (with F the remaining focus).

Of course, this is consistent with the polar form: the eccentricity satisfies $e = c/a = 1$ for a parabola. One can similarly think of the parabola as an approximation/limit of hyperbolas as $a, c \to \infty$ while keeping $(c - a)$ constant.

As the eccentricity of an ellipse tends to 1, the second focus recedes to ∞ along the axis; in the limit it becomes a remote focus. Hence the "directrix circle" construction from the ellipse case no longer survives as a literal circle centered at the observed focus. Yet the effect of that construction does survive: a circle of infinite radius, centered at the remote focus, degenerates into a straight line perpendicular to the axis—namely the parabola's directrix. In this limiting viewpoint the directrix may be regarded as the remnant of the former directrix circle, positioned so that it meets the axis AF at a point symmetrically placed with respect to the vertex A (in the same sense as in the elliptical construction). So while an ellipse and a hyperbola each have two directrix circles and two directrix lines, a parabola has only one observable directrix "circle" and one directrix line; furthermore, this circle degenerates to the same line.

However, we will also show that a parabola still admits several natural associated circles that encode useful geometry. In particular, although the earlier auxiliary circle and directrix circle have both degenerated into a line, one may still construct a hodograph circle with various scalings, similar to what we have done for the ellipse and hyperbola, which remains genuinely circular and retains the kinematic/geometric information needed for the argument.

In this section we show that the analogues of Proofs 2 and 3 from the ellipse case also hold for the parabola, with some special care. We recommend reading Newton-style arguments (in the 1687 tradition), such as the historical collection of various properties of the parabola in [Sta19]. However, a much clearer explanation of the hodograph for the parabolic case is given by Derbes (2001) [Der01], which uses the circle centered at the focus F with radius $2a$. With this circle, the Maxwell (1877)/Hamilton (1847) hodograph proof for the direct problem extends naturally to the parabola.

7.3.2 Parabola variant of Proof 2: a constant BR^2/BD

This is the parabola analogue of Proof 2 in the auxiliary-circle style.

Proposition 7.1 (Parabola variant of Proof 2)**.** *In the parabolic construction (notation matching the ellipse case), as $B \to A$ one has*

$$\boxed{\lim_{B \to A} \frac{BR^2}{BD} = 4a,} \tag{7.2}$$

where a is the parabola parameter (so the semi-latus rectum is $p = 2a$).

Proof. As $B \to A$ (small angles $\delta, \delta' \to 0$), similar triangles in the figure give

$$\frac{A'B'}{BD} = \frac{B'F}{DS} \sim \frac{2\,B'F}{DA}. \qquad (1)$$

Resolve the ratio with the common angle α:

$$\frac{B'F}{DA} = \frac{B'F \sin\alpha}{DA \sin\alpha} \sim \frac{a}{BR}. \qquad (2)$$

since $DA \sin\alpha \sim BR$ and $B'F \sin\alpha = a$. Substituting (2) into (1),

$$\frac{A'B'}{BD} \sim \frac{2a}{BR}, \qquad \text{so} \qquad BD \sim \frac{A'B'\,BR}{2a}. \qquad (3)$$

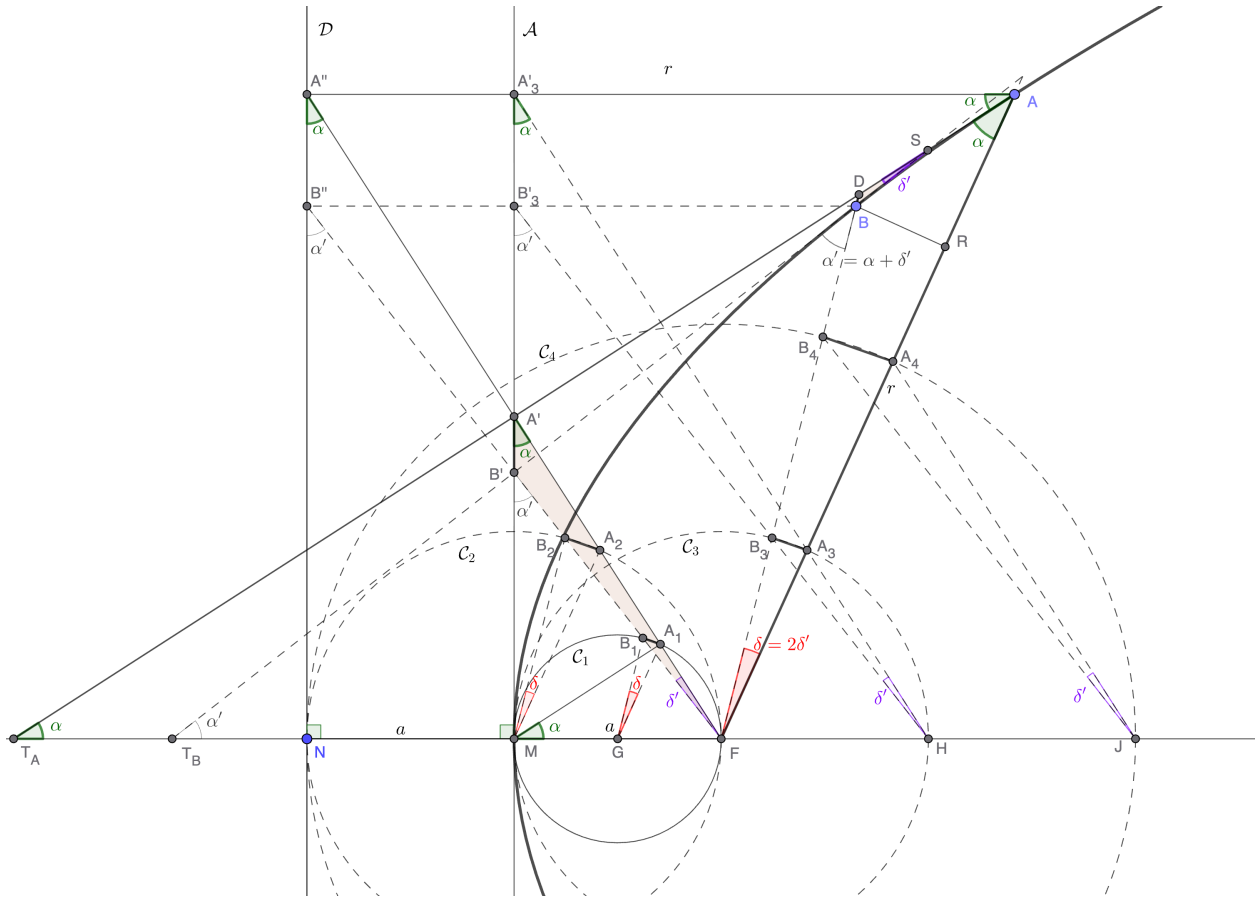

Figure 6: Parabola construction used in Section 7.3 (parabola analogue of Proof 2).

Also $A'B' \sim r\delta'$ and $BR = 2r\delta'$, hence $A'B' \sim \frac{1}{2}BR$. Therefore

$$BD \sim \frac{(\frac{1}{2}BR)\,BR}{2a} = \frac{BR^2}{4a},$$

so $\dfrac{BR^2}{BD} \to 4a$.

An equivalent shortcut (as suggested) is to use $\triangle SDB \sim \triangle FDS$, giving

$$DS^2 \sim DB \cdot FS, \qquad DB \sim \frac{DS^2}{FS}.$$

With $DS = \frac{1}{2}AD$ and the same α-projection relations, one again obtains

$$DB \sim \frac{BR^2}{4a},$$

hence the same limit $\dfrac{BR^2}{BD} \to 4a$. $\square$

Theorem 7.1 is the direct parabolic counterpart of equation (6.10) in the elliptic setting. Since

$$\frac{BR^2}{BD} \to 4a \quad \Longleftrightarrow \quad \frac{BD}{BR^2} \to \frac{1}{4a},$$

Theorem 3.2 gives

$$a_F = \frac{2L^2}{r^2} \cdot \frac{1}{4a} = \frac{L^2}{2a}\,\frac{1}{r^2} = \frac{L^2}{p}\,\frac{1}{r^2} \qquad (p = 2a),$$

so the same inverse-square law follows.

8 Reverse Problem Discussion

8.1 Hodograph approaches and historical remarks

8.1.1 Newton, uniqueness, and asserted conics

From the *Principia* viewpoint (1687), Newton gives a geometric proposition-chain establishing both directions: conic with force at a focus ⇒ inverse-square law, and inverse-square centripetal attraction ⇒ conic orbits (ellipse/parabola/hyperbola by regime) [New87, Cha95]. What he does *not* provide is a modern initial-value existence/uniqueness theorem for the inverse problem; the argument is synthetic and limit-geometric (ultimate ratios), not an ODE well-posedness proof.

In parallel, early Continental analysts (late 17th to early 18th century) reduced the central-force problem by $u(\theta) = 1/r$, yielding Binet's equation. For $F(r) = -\mu/r^2$:

$$u''(\theta) + u = \frac{\mu}{h^2}, \tag{8.1}$$

with solution

$$u(\theta) = \frac{\mu}{h^2}\Big(1 + e\cos(\theta - \theta_0)\Big), \tag{8.2}$$

i.e. a conic in polar form. However, deriving this form by itself is not a substitute for a uniqueness theorem: without an existence/uniqueness framework, one has not yet formalized exclusion of other possible local branches/continuations. Historical accounts of this transition to differential methods are discussed by Nauenberg (2003) [Nau03].

So the modern statement "initial data determine a unique orbit" is best read as a reconstruction of Newton's practice rather than a theorem he states in contemporary form. The same caution applies to early Continental differential reductions: the orbit equation gives candidate solution families, while later uniqueness theory is what formally secures single-orbit determinacy from initial data. Likewise, hodograph language is absent in the *Principia* (1687) and appears later with Hamilton (1847). Rigorous local existence/uniqueness theorems for ODEs were developed much later (Cauchy in the 1820s; then Lipschitz/Picard–Lindelöf in the late 19th century, roughly the 1870s–1890s), i.e. more than a century after 1687.

8.1.2 Hamilton's circular hodograph for the inverse-square law

Hamilton introduced the hodograph in 1847 as the curve traced by the tip of the velocity vector $\mathbf{v}(t)$ in velocity space, and showed that for motion under a central inverse-square force the hodograph is a circle [Ham47].

Write the dynamics as

$$\ddot{\mathbf{r}} = -\frac{\mu}{r^2}\,\hat{\mathbf{r}}, \qquad h := r^2\dot{\theta} = \text{const}.$$

Reparametrize by θ:

$$\frac{d\mathbf{v}}{d\theta} = \frac{d\mathbf{v}}{dt}\frac{dt}{d\theta} = \ddot{\mathbf{r}}\,\frac{r^2}{h} = -\frac{\mu}{h}\,\hat{\mathbf{r}}. \tag{8.3}$$

Using the polar-frame identity

$$\frac{d\hat{\boldsymbol{\theta}}}{d\theta} = -\hat{\mathbf{r}},$$

equation (8.3) becomes

$$\frac{d\mathbf{v}}{d\theta} = \frac{\mu}{h}\frac{d\hat{\boldsymbol{\theta}}}{d\theta}. \tag{8.4}$$

Integrating once gives

$$\frac{d}{d\theta}\Big(\mathbf{v} - \frac{\mu}{h}\hat{\boldsymbol{\theta}}\Big) = 0 \quad \Longrightarrow \quad \mathbf{v} = \mathbf{C} + \frac{\mu}{h}\hat{\boldsymbol{\theta}}. \tag{8.5}$$

with constant vector $\mathbf{C}$. Because $\|\hat{\boldsymbol{\theta}}\| = 1$,

$$\|\mathbf{v} - \mathbf{C}\| = \frac{\mu}{h}.$$

Hence the hodograph (the tip of $\mathbf{v}$ in velocity space) is a circle of radius μ/h, centered at $\mathbf{C}$. Later expositions (notably Maxwell, 1877) emphasize that the hodograph circle is not only a consequence of inverse-square attraction, but also a practical route toward reconstructing the orbit and diagnosing the force law when combined with angular momentum conservation [Max77, Der01]. In this pedagogical tradition (Hamilton, 1847 → Maxwell, 1877 → Feynman, 1964, with later refinements), one studies both Kepler problems: direct (force ⇒ orbit) and inverse (orbit laws ⇒ force) [Der01].

8.1.3 Feynman's Lost Lecture, strengths, and later repairs

Feynman's 1964 lecture re-popularized the hodograph method for teaching: first establishing a circular hodograph for inverse-square attraction, then reconstructing the conic orbit geometrically [GG96]. Later work, including Derbes (2001), clarified missing details in tangent placement and invariants; in particular, angular momentum conservation is the extra ingredient that fixes line placement and scale [Der01]. More recent refinements, including van Haandel–Heckman (2009) and follow-up work by Cariñena–Rañada–Santander (2016), may be read as using either (i) the directrix circle centered at the second focus, or (ii) an equivalent directrix-circle/hodograph picture centered at the force center, with a $\pm 90°$ rotation between velocity and geometric proxy [vHH09, CnRnS16]. CNRNS16 emphasizes that the force-center version gives a cleaner Feynman-style transformation.

In our constructions (Figures 5 and 6), the auxiliary circle has the same dual role. In particular, from the local identity used in Theorem 6.1,

$$FH \cdot FK' = b^2, \qquad F'A' = FK',$$

so $\overrightarrow{F'A'}$ and $\overrightarrow{FK'}$ are colinear with opposite orientation and equal magnitude. Hence one can use the auxiliary-circle data with either center choice (F or F') as a hodograph proxy, corresponding to a $+90°$ or $-90°$ rotation. For the parabola, the same logic yields four equivalent rotated-hodograph constructions; Derbes' choice is one of them (his C_4 circle). The remaining step is the energy classification: $E < 0$ gives an ellipse, $E = 0$ the parabolic limit, and $E > 0$ a hyperbola.

8.2 Our approach (Newton-style discretization built on the hodograph circle)

We take as given (from Section 8.1) the hodograph-circle fact: for planar motion under a central inverse-square force, the hodograph of $\mathbf{v}(t)$ is a circle in velocity space.

Our goal is to recover the orbit form $r(\theta)$ by a finite-step argument aligned with Newton's impulse-triangle logic, minimizing reliance on continuous calculus.

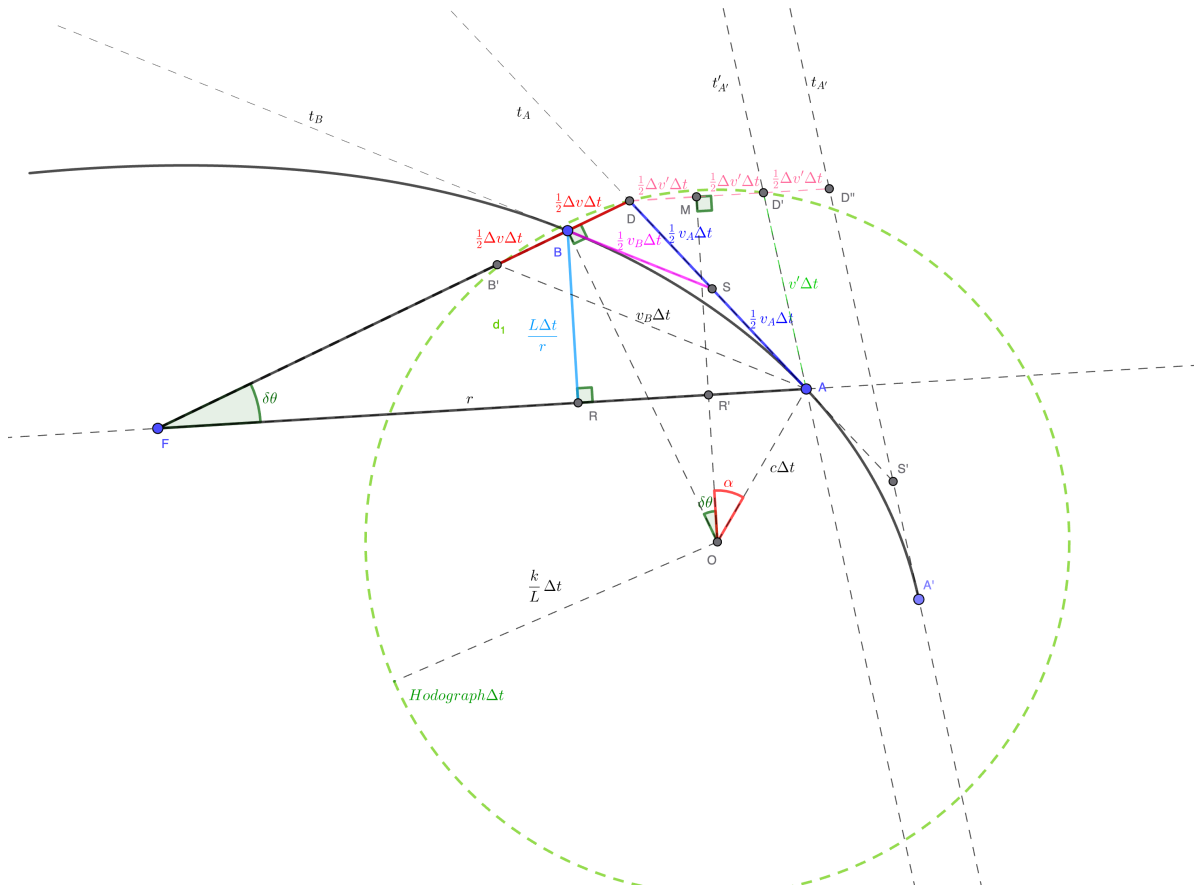

Figure 7: Inverse-problem construction in Section 8.2.1: previous/current/next velocities $(V_{A'}, V_A, V_B)$, each scaled by Δt, are shown on the translated Δt-scaled hodograph circle, with decomposition $OM = BR + OR'$.

8.2.1 Hodograph-radius analysis and translated-circle reconstruction

Assume planar motion under a centripetal acceleration toward F:

$$a = \frac{k}{r^2}, \qquad r = FA, \tag{8.6}$$

and constant specific angular momentum

$$L = r^2\dot{\theta}. \tag{8.7}$$

Fix a small time step Δt, with consecutive points $B \to A$ on the orbit, $BR \perp FA$, and the translated Δt-scaled hodograph geometry in which

$$OM = BR + OR'. \tag{8.8}$$

Define

$$u := \frac{k}{L}. \tag{8.9}$$

Proposition 8.1 (Hodograph-radius analysis $\Rightarrow$ conic form)**.** *With the notation of Figure 7, the hodograph of* $\mathbf{v}$ *is a circle of radius* u *(hence the* Δt*-scaled hodograph circle has radius* $u\Delta t$*). If the fixed shift satisfies* $OR' = c\,\Delta t\cos\alpha$*, then*

$$\frac{1}{r} = \frac{k}{L^2} - \frac{c}{L}\cos\alpha \quad \Longleftrightarrow \quad r(\alpha) = \frac{p}{1-\varepsilon\cos\alpha}, \qquad p = \frac{L^2}{k}, \quad \varepsilon = \frac{cL}{k} = \frac{c}{u}. \tag{8.10}$$

Moreover,

$$\begin{aligned}\frac{c}{u} < 1,\ = 1,\ > 1 &\iff \textit{velocity-origin inside/on/outside hodograph circle}\\ &\iff \textit{ellipse/parabola/hyperbola.}\end{aligned} \tag{8.11}$$

Proof. In Figure 7, the previous, current, and next velocities $(V_{A'}, V_A, V_B)$ are placed together in velocity space after scaling by Δt; this is exactly the translated Δt-scaled hodograph circle used in the relations below.

First, Δv depends only on $\Delta\theta$. Over a short interval Δt,

$$\Delta v = a\,\Delta t = \frac{k}{r^2}\Delta t.$$

From $L = r^2\dot{\theta}$, one has $\Delta t = \frac{r^2}{L}\Delta\theta$. Hence

$$\Delta v = \frac{k}{r^2}\cdot\frac{r^2}{L}\Delta\theta = \frac{k}{L}\Delta\theta = u\,\Delta\theta, \qquad \frac{\Delta v}{\Delta\theta} = u \text{ (constant)}. \tag{8.12}$$

Next, conclude the hodograph-circle radius. Because the force is central, $\Delta\mathbf{v}$ is radial inward. As FA turns by $\Delta\theta$, the direction of $\Delta\mathbf{v}$ turns by the same $\Delta\theta$, while its magnitude is $u\,\Delta\theta$ by (8.12). Thus the velocity-tip polygon is the equal-angle, equal-arc-length limit of an inscribed circle polygon, so the hodograph is a circle of radius u. Multiplying by Δt gives

$$OM = u\,\Delta t. \tag{8.13}$$

Then, use tangential displacement. The tangential speed is $v_\theta = L/r$, hence

$$BR = v_\theta\Delta t = \frac{L}{r}\Delta t. \tag{8.14}$$

Now use the key geometry in the plot. From the translated-circle construction, $OM = MR' + OR'$ and $MR' = BR$, so

$$OM = BR + OR'. \tag{8.15}$$

Assume

$$OR' = c\,\Delta t\cos\alpha. \tag{8.16}$$

Substitute (8.13), (8.14), (8.16) into (8.15):

$$u\Delta t = \frac{L}{r}\Delta t + c\Delta t\cos\alpha.$$

Cancel Δt:

$$u = \frac{L}{r} + c\cos\alpha \quad\Longrightarrow\quad \frac{1}{r} = \frac{u}{L} - \frac{c}{L}\cos\alpha = \frac{k}{L^2} - \frac{c}{L}\cos\alpha.$$

This is equivalent to

$$r(\alpha) = \frac{p}{1-\varepsilon\cos\alpha}, \qquad p = \frac{L^2}{k}, \quad \varepsilon = \frac{cL}{k} = \frac{c}{u}.$$

Finally, classify by the origin position in velocity space. The hodograph circle has radius u, and the velocity-origin offset has magnitude c. Hence the origin is inside/on/outside the hodograph circle according as $c < u$, $c = u$, $c > u$, i.e. $c/u < 1, = 1, > 1$. Since $\varepsilon = c/u$, these are precisely $\varepsilon < 1, = 1, > 1$, giving ellipse/parabola/hyperbola. □

9 Summary and Future Work

This paper develops a Euclidean, *Principia*-style treatment of the Kepler problem using finite-step constructions, tangent/triangle geometry, and Newton's ultimate-ratio viewpoint. For the direct problem, the proof chain across Sections 3–7 is now explicit and modular. Section 3 establishes the dynamical bridge: from constant areal speed, the acceleration direction is central, and the two local lemmas convert a local geometric ratio into the inverse-square force form. Section 4 sets the geometric transport layer by introducing the auxiliary circle and affine map, so tangent and parallel structures can be moved cleanly between circle and ellipse. Section 5 executes the core local-drop computation in this transported geometry: tangent transfer, matching normal drops, and swept-area transport reduce the local deflection to a conic-invariant ratio, and Section 5.7 is stated as Proof 1 for the ellipse. The same architecture is used throughout: conic-specific geometry computes the local ratio, while the Section 3 conversion step remains universal. Section 6 then gives Proof 2 and Proof 3 as independent auxiliary-circle derivations, confirming the same force law from different local identity chains and clarifying the hodograph-facing interpretation. Section 7 applies the same architecture to the non-elliptic conics: hyperbola and parabola are treated as geometry replacements in the local-ratio step, with the same final conversion to the inverse-square law.

For the reverse problem (Section 8), we keep the hodograph-circle theorem in view but work in a discrete Newtonian style. The key construction is the translated hodograph scaled by Δt and carried along the orbit: previous, current, and next velocities are placed together in the same small-circle picture, and the corresponding tangent/offset geometry is used to reconstruct the conic form in configuration space. This gives a diagrammatic bridge from velocity-space circularity to orbit-shape recovery without relying on a continuous differential-equation derivation at each step. The main limitation remains uniformity of local geometry across conic types: the elliptic affine-circle method is especially clean, while parabola and hyperbola require additional case-specific constructions. The next step is therefore to package Sections 5–7 into a single conic-uniform local-drop framework, so that one canonical geometric template handles all three energy regimes while preserving the same Section 3 conversion principle and the same translated, Δt-scaled hodograph interpretation in the reverse direction.

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